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1 draft: text capture

1.1 sketches

1.1.1 data capture -> vulnerability

In the process of capture, what meaning or knowledge is lost or inextractible?

- *Data Grab*: When these plants are taken and commercialized by the colonizer? It's a question of code as translation and commercialization.
 - there is something that doesn't translate in these knowledge practices, something to do with relation, community.

Chun: what makes a user vulnerable with regard to their language? In what ways is web scraping/capture making people vulnerable? Is there anything that is opened up by this vulnerability? How does language on the web express this vulnerability?

- there is something being taken from us when we are online.

What good can come from being vulnerable? What kind of resilience? Being online means exposure, exploitation, extraction.

"A certain loss of control drives communication, drives our using" (76)
- there is always a give and take when information is put out there. The question, for this age of capture, what is our state of vulnerability now?

1.1.2 outline

- intro
 - with each passing day, it becomes harder and harder to scrape the web.
- web scraping
 - how is text scraped?
 - blockers
 - * circumventing blockers
- the politics of capture
 - the forces which extract for exploitation (Couldry & Mejias, Zuboff, Chandrasekhar);
 - ☒ Mejias & Couldry: from coercion to consent
 - the technical strategy is to leverage HTTP data.
 - the legal strategy relies on users not knowing the extent of their rights being taken.
 - ☐ the effect on web scraping (Chandrasekar, Cloudflare)
 - ☐ Zuboff: "logic of accumulation"
 - the concept of informate, the emergence of new data, new realms of extraction.
 - value is created from value. Value finds a way to perpetuate itself.
 - ☐ today's obsession with scaling up (Bender et al, Varoquaux et al)
 - creates a new territory for extraction (Cloudflare)
- how to resist capture? close reading.
 - thinking small
 - * masking size/capital for innovation (Whittaker)
 - * the rise of chinese models Deepseek and Zimi
 - * the performance limits of scale (Varoquaux, *Slow Down*, Moore)
 - reading for vulnerability/exposure

- * there is a vulnerability that drives communication (Chun)
 - "vulnerability ... drive[s] communication" (76).
 - "potential ... from vulnerability" (297).
 - * exposing the self as a method of healing (Anzaldua)
 - reading for vulnerability is a kind of reading that cannot be automated
 - * it requires close and individualized attention
 - * something is always lost in communication (Schutte)
- scraping protocol
 - writing crawlers to scrape ACLU & Heritage Foundation.
 - crawlers employ recursion to gather all of the articles.
- close reading
 - language on gender across polarized views
 - * the phantasm of gender ideology (Butler)
 - reading the datasets for traces of this ideology
 - * patterns around the terms reality and subjectivity
 - * point to trans materialist argument – about making gender explicit, expressive.
 - there is a parallel between the expressiveness of language and of bodies – this parallel is what creates a rift between queer and trans studies, a rift I picked up in chapter 5.
- the legal route:
 - the legal debate over fair use (court cases, "Comments" on copyright)
 - * web scraping creating new legal paradigms (Cohen)
- solutions:

- small, local solutions
 - * ("Losing the API", Mohammed et al)
 - * allowing one to bypass blockers
 - * scraping congress.gov
- collective approaches to data
 - * communal conceptions of data ownership and stewardship (African NLP, Grohmann)
 - * communal conceptions of publication through production (Moore)

1.1.3 argument

How can we re-work capture practices to resist the co-option of meaning for exploitation and extraction?

Meaning is being co-opted across various scales. On the scale of individual words, like "democratize" and "soverignty" in tech discourse and policy, and also phrases like "protect women" in gender discourse. But also on the large scale of data extraction. While there has always been a form of extraction that drives capitalist systems, this extraction of language at scale is new.

How can the protocols of extraction be re-worked to resist it?

- reworking via counterdata, creating counter datasets.
- reworking via close reading.
 - reading for vulnerability. for what is exposed. for fear.

1.2 draft

1.2.1 introduction

There is a co-option of meaning occuring today.

1.2.2 web scraping

1. what is web scraping?

(a) from bytes to HTML

Web scraping is the extraction of information from websites. It most often involves (with some exceptions, explained below) working with data in HTML format, with elements and attributes.

```
<!DOCTYPE html>
<html>
<body>

<h1 class="header">First level heading element</h1>

<p>A paragraph element with a <a href="/">link</a> inside of it</p>

</body>
</html>
```

Websites are stored in web servers, accessed over a network. Web data travels over the network in the form of bytes, the smallest digitally addressable data unit. When the bytes reach the web browser, they are re-encoded into HTML language, which is assembled into a web page readable by human eyes.

(b) HTTP

Web *scraping* is possible by manipulating HTTP, or HyperText Transfer Protocol, which is the language of the network. A web scraper uses code to manipulate an HTTP request so that it receives the website data in the form of code.

Web scraping is, at its core, about mimicking a typical browser request. Using HTTP protocols, it makes a request to a web server much like a user makes a request by typing in a URL into the browser. The data it gets back from the HTTP request is also the same as the data that comes to a browser after searching a URL—only the data is intercepted before in the form of bytes, rather than assembled into a web page visible through a browser.

```
r = requests.get("https://www.nytimes.com/")
```

You can scrape a website with only five lines of code. Below is an example that scrapes the home page of *The New York Times* website using the **bs4** and **requests** libraries in python.

```

from bs4 import BeautifulSoup
import requests

r = requests.get("https://www.nytimes.com/")
source = r.content
soup = BeautifulSoup(source, 'html.parser')

```

The `requests` library is one of the most popular libraries in the python programming language. Enabling programs to send and receive data through HTTP networks (i.e., the internet), it is the bedrock for virtually all web scraping and API tasks. The `get()` function from `requests` can access the *NYTimes* homepage, which is saved in the form of bytes via the `content` property. Then, using `bs4`, a python library for working with HTML data, the data is parsed into an object called `soup` (following `bs4`'s naming conventions), which is a pythonic object of HTML data.

The beauty of `bs4` is that it makes HTML—a notoriously fussy language that is the reason why so many web frameworks exist in the first place, attempting to abstract the user from working directly in HTML—accessible through simple python syntax.

For example, the expression `soup.title` returns the following output:

```

>>> soup.title
<title data-rh="true">The New York Times - Breaking News, US News, World News

```

Similarly, other HTML elements can now be accessed using python dot syntax.

```

>>> soup.p
<p>Make sense of the day's news and ideas.</p>

>>> soup.a
<a class="css-kgn7zc" href="#site-content">Skip to content</a>

>>> soup.h1
<h1 class="css-1dv1kvn">New York Times - Top Stories</h1>

```

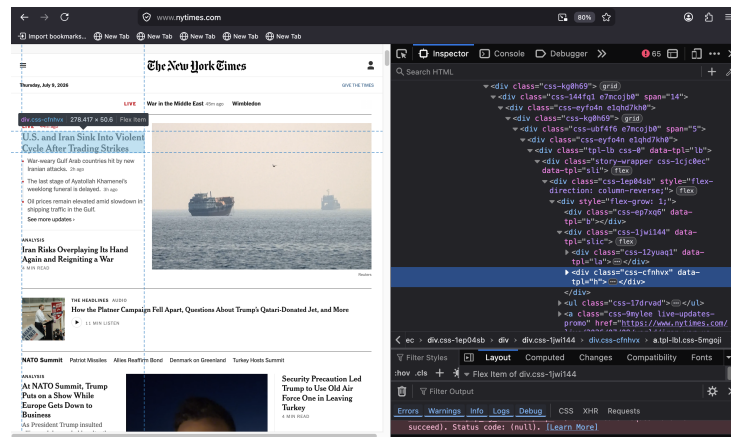
Further properties like `text` (for just text) and functions like `find()` are also possible. Functions can be further specified when combined with the `class_` parameter, to search the HTML with more precision. Due to python syntax, they can also be chained together to create complex expressions. These building blocks enable programmers to navigate the HTML tree with granularity.

```
>>> soup.h1.text
'New York Times - Top Stories'

>>> soup.find('div', class_ = 'css-cfnhvx')
<div class="css-cfnhvx" data-tp1="h"><a class="tp1-lbl css-5mgoji" data-tp1="

>>> soup.find('div', class_ = 'css-cfnhvx').text
'U.S. and Iran Sink Into Violent Cycle After Trading Strikes'
```

HTML classes for an element can be found by using built-in browser tools like the web inspector. On the webpage, right click on the relevant text and select **inspect** from the dropdown menu. The element will be highlighted with information for the `class` attribute.¹



¹Note that across programming languages like HTML and python, terms like "attributes" mean different things and in practice function very differently. In HTML, an attribute is data contained within an HTML element, like a value specifying a CSS class, such as `<h1 class=title>New York Times</h1>`. By contrast, in python an attribute is information associated with an object model, like `r.content` or `r.status_code`, which is data associated with an HTTP request object, `r`.

Using `requests`, `bs4`, and the web inspector, one can easily grab a list of the current headlines from the *NYTimes* homepage.

```
>>> for i in soup.find_all('div', class_ = 'css-cfnhvx'):
    print(i.text)
U.S. and Iran Sink Into Violent Cycle After Trading Strikes
Iran Risks Overplaying Its Hand Again and Reigniting a War
At NATO Summit, Trump Puts on a Show While Europe Gets Down to Business
Security Precaution Led Trump to Use Old Air Force One in Leaving Turkey
'A Slow-Rolling Disaster': Inside the Implosion of the Platner Campaign
What's Next for Democrats in the Senate Race in Maine
How Democrats and Republicans Diverge on Sexual Assault Allegations
Platner Suspends Senate Bid in Maine After Rape Accusation
Trump's Loyal Defender at the Vatican
Trump Says He'll Ask Supreme Court to Rehear Citizenship Case, an Unlikely Event
Can Trump's Arch Be So Tall? A Law May Be Redefined to Get to Yes.
Morocco Faces France and Its Star, Mbappé, in Quarterfinals
A Colonial Past Mixes With the Global Present
How Morocco's Diaspora Became the Team's Driving Force
13 Red Cards Have Been Given at the World Cup. Not All Have Been Equal.
```

Anything else to say about HTTP?

(c) Blockers and the resistance

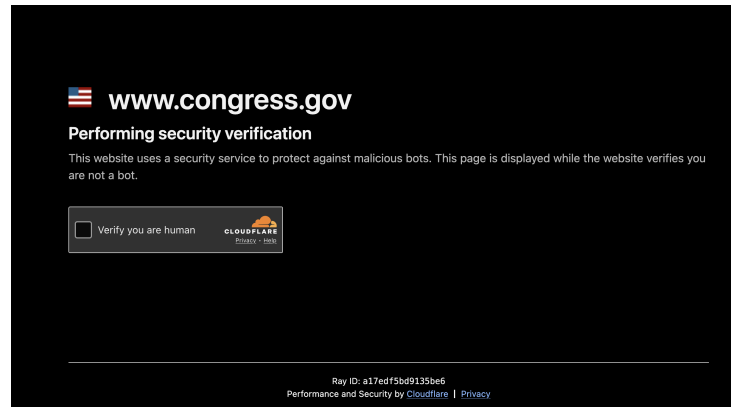
Web servers, unfortunately for the scraper, often block automated requests to their data. They use various forms of "blockers" or mechanisms that deny the HTTP request, sending back a code 403, or "Forbidden" message.

Each year of the past few years, bot traffic has increased exponentially, coming mostly from major tech and AI companies, especially OpenAI, META, Google, and Anthropic. According to Cloudflare, 2025 saw increases of 19% in bot traffic (See Fig #) (Cloudflare, "From Googlebot to GPTBot: who's crawling your site in 2025"). It is no surprise that traffic has increased mostly from these companies, whose ML model training requires ever increasing amounts of data.

Rank	Bot Name	Share (May 2024)	Rank	Bot Name	Share (May 2025)
1	Bytespider	42%	1	GPTBot	30%
2	ClaudeBot	27%	2	ClaudeBot	21%
3	Amazonbot	21%	3	Meta-ExternalAgent	19%
4	GPTBot	5%	4	Amazonbot	11%
5	Applebot	4.1%	5	Bytespider	7.2%

As automated scrapers increase in frequency, so do the bot blockers. Over the past two years, Cloudflare, which runs on 20% of total internet pages and 50% of pages based in the US, has added increasing bot detection like "Challenges" and "Turnstiles" to its web pages. Some of these operate under the surface and virtually invisible to the user, such as checking the HTTP request meta-data like the "User-Agent." By default, the User-Agent is set to the name of web browser and operating system making the request. For example, my Firefox browser running Mac OSX has the User-Agent `Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:152.0) Gecko/20100101 Firefox/152.0`. The `python requests` library, by contrast, sets the User-Agent to `python-requests/2.32.3`, making any HTTP request using that library easy for blockers to detect.

Beyond examining HTTP request data, bot blockers will use JavaScript code to check if the request is from a real web browser, rather than from an automated program. JavaScript, which is a programming language native to web browsers (and discussed in more detail in Chapter 4: Text Display), can inject challenges for the browser to solve, challenges which may or may not involve the user. All of these challenges might require some processing from the browser, and increasingly, they may require the user's interaction (to confirm that the request comes from a real browser, not a bot). One popular challenge is the "security verification" checkbox, which is increasingly visible across the internet (See Fig #).



While these kinds of blockers are more advanced, there are ways of overcoming them. One effective method involves using a virtual or "ghost" browser through python libraries like `selenium`. Instead of sending an HTTP request via a python script, `selenium` routes the request through an actual web browser—though one that is operated remotely by the python script. When the program is running, a browser window will open on its own, navigate to a web page, and click on links, all without the input of the user. It looks like a "ghost" is operating the computer. One major drawback to this method is that it requires a virtual display, which many "headless" scraping servers do not have, in order to access the data via the browser window. That requirement, in addition to the increase in necessary processing power, detracts large-scale web crawling.

At the same time, internet security companies like Cloudflare have also moved to what they call a "permission-based model," where bots must pay for seamless access to certain domains. A blog post frames the new policy as the next logical step in the mass data extraction environment.

For decades, the Internet has operated on a simple exchange: search engines index content and direct users back to original websites, generating traffic and ad revenue for websites of all sizes. This cycle rewards creators that produce quality content with money and a following, while helping users discover new and relevant information. That model is now broken. AI

crawlers collect content like text, articles, and images to generate answers, without sending visitors to the original source – depriving content creators of revenue, and the satisfaction of knowing someone is viewing their content. (“Cloudflare Just Changed How AI Crawlers Scrape the Internet-at-Large; Permission-Based Approach Makes Way for A New Business Model”)

The argument refers to the original purpose of large-scale web crawling, which was to populate and otherwise assist search engines. This activity of taking content in the form of search results was legitimized in a famous copyright lawsuit against Google, *The Authors Guild Inc., et al. v. Google, Inc.* (2015). In this case, the judge ruled that search results, such as that from Google Books, were considered "fair use" because they point users to the original source without diminishing its value. As long as search engines direct traffic to the original source, then there appears to be no infringement.

Cloudflare points out that today’s scrapers and crawlers do not direct users back to the original source; rather, the AI training process *severs* the data completely from its context, including its source.² Web pages are taken from their context and compiled into massive collections which are fed, bit by bit, into a machine learning algorithm. The severed connection to the source cannot be monetized, "depriving content creators of revenue... and satisfaction."

As a solution, Cloudflare is shifting their policy to protect media outlets and publishers. They have begun to offer the "permission based model," where companies must pay content creators for access to their web data. If the scraper does not have explicit permission to retrieve data, then they will be blocked. The announcement of Cloudflare’s new policy came with written endorsement from publishers and media outlets. Roger Lynch, CEO of media conglomerate Condé Nast, which owns *Vogue*, *The New Yorker*, *Architectural Digest*, *Vanity Fair*, *Pitchfork*, *Wired*, *Bon Appétit*, *Ars Technica*, among others, declares that, "Cloud-

²As I explain in the next chapter on "Text Analysis," deep learning methods for working with language data essentially chop up text into computable units, that is, into words. They take words one by one, called "tokens," to study the relationships between them.

flare's innovative approach to block AI crawlers ... is a critical step toward creating a fair value exchange on the Internet that protects creators, supports quality journalism and holds AI companies accountable" ("Cloudflare Just Changed How AI Crawlers Scrape the Internet-at-Large; Permission-Based Approach Makes Way for A New Business Model"). Of course, there is no guarantee (or even expectation) that such payments would reach the pockets of actual content creators, authors, copywriters, artists, who are populating these sites with their work. If the new policy rewards someone, its not the original creator but the publisher who enters into agreements with the tech companies. In that sense, the "fair value exchange" cited by Lynch is in fact another way of further monetizing what is already an extractive practice of capture.

1.2.3 the politics of capture

1. the current data grab We are in the midst of an intense acceleration of what Ulises A. Mejias and Nick Couldry call a "data grab." The data grab is the heir of colonialism's "land grab," the long process over several centuries in which Europe, despite representing 8 percent of the world's landmass, seized over 84 percent of its total territory (Mejias & Couldry 5). According to Mejias and Couldry, this data grab took traditional colonial processes of exploration, extraction, and exploitation of resources and applied them to a new object, to data. "[B]oth systems," they explain, "are founded on an appropriation of the world's resources, treating them as 'just there' for the taking. Historically it was the territories. . . In the contemporary era, it is human life, extracted in the form of data" (Mejias & Couldry 18).

The object, and particularly its digital nature, required new strategies for capture that differ from those of land-based resources. First, rather than partition zones of extraction in the form of land territories, there has emerged the digital platform. The platform is a digital space, which can range in scope from individual websites on the internet to self-enclosed apps, which contains mechanisms that gather up details of the user's identity and activity—not only that which has been voluntarily given, but also that contained in hidden codes and protocols that append all data traveling over the internet. In more technical terms, the platform is a server location, from which the user requests data (via the HTTP protocol), and in that request, also sends

some information to the server. Most recently, the best way of tracking users is in the form of a "cookie," a bit of data which the server assigns to the user and which can be used to track that user's activity on that and other servers. (Although, cookies can also be manipulated by users to simulate real web requests that can bypass web blockers, as I demonstrate at the end of the chapter).

If the first strategy takes advantage of technical protocols, the second strategy takes advantage of legal ones. Besides new procedures for capture, it is necessary also to create the proper pretense for it. In traditional colonialism, Mejias and Couldry explain, this pretense was explicitly coercive. They offer the example of the *Requerimiento*, a document that the Spanish colonizers read to the indigenous peoples of Americas as they entered their land and villages. The document certified the Spanish authority, in the name of the king and the church, to take the land, its resources and people, under Spanish control. It declared all who heard it subject to the crown, and any those who resisted to be violently subdued. Mejias and Couldry compare this document to the "Terms of Service" agreement from Google Chrome, which also makes claims upon the user, in particular on the "Content" that a user inputs into the system. The 2007 version of the TOS asserts that,

You give Google a perpetual, irrevocable, worldwide, royalty-free, and non-exclusive license to reproduce, adapt, modify, translate, publish, publicly perform, publicly display and distribute any Content which you [the user] submit, post or display on or through, the Services. (Rpt. in Mejias & Couldry 12).

More recently, the wording of TOS has muted the explicitness of this language. Recent wordings of Google TOS, the authors point out, don't need to be as assertive: "without most of us ever having read those original terms and conditions. . . . the extraction of data is already well underway" (Mejias & Couldry 12). The more recent strategy has shifted from assertive to excessive and at times obscure language (these TOS now occupy many pages). For example, the 2024 Google TOS, which is now held on its own subdomain, <https://policies.google.com>, phrases like "operating and improving the services" and "to customize our services for you" offer the necessary cover for "using automated systems and algorithms to analyze your content" ("License", "Google

Privacy and Terms"). Additionally, the TOS contains strange and difficult to parse formulations, like "We need your permission if your intellectual property rights restrict our use of your content. You provide Google with that permission through this license" ("License", "Google Privacy and Terms"). The meaning appears to be that Google will need, yet already has obtained, permission from the user. But what exactly is one giving permission for Google to do? The TOS does not say.

That is if one even bothers to read the terms of service. Whereas the *Requerimiento* relied on blunt language claiming an authority over land and people, more recent TOS rely on the user not even checking the terms. While these approaches, separated by centuries, differ dramatically, they both rely on the user not knowing the extent of the rights being taken from them. The *Requerimiento*, similar to modern-day TOS, used a language that was obscure to its audience. As Mejias and Couldry point out, "[the conquerors] read this document in Spanish. . . to indigenous or First Nation peoples who did not speak Spanish!" (13). The legal strategy, then, maintains this dependence on a certain level of ignorance from the user.

While Mejias and Couldry discuss present-day data extraction with regard to platforms—bounded tools, apps, or websites controlled by tech companies like Alphabet or META—there is another major extraction underway, which is in spirit closer to the one with land. That is the major extraction occurring via AI-powered web crawlers. According to Cloudflare, a major tech provider for cloud and cyber computing services, AI crawlers increased sharply, ranging between 18%-48% in the months between May 2024 and May 2025. From this time period, the highest increases came from OpenAI, Claude, and Google's AI crawlers.³

This situation has created both concern and excitement. As researcher Ramya Chandrasekhar points out, "certain well-resourced actors like Big Tech are able to take advantage of data flows on the open web to internet to train proprietary foundation models, while giving little to no value back to either the maintenance of shared informational resources or communities of commoners" (Chandrasekhar 2).⁴

³"From Googlebot to GPTBot: Who's Crawling Your Site in 2025." The Cloudflare Blog, 1 July 2025, <https://blog.cloudflare.com/from-googlebot-to-gptbot-whos-crawling-your-site-in-2025/>.

⁴Chandrasekhar, Ramya. Legal Frictions for Data Openness: Reflections from a Case-

Meanwhile, on the side of industry specialists, like the company Zyte, the outlook is positive. The company's annual report embraces the increasing of web scraping, even describing it in inevitable terms:

For years, we have talked about the "cat-and-mouse" between data-gatherers and data owners. But 2025 marked a significant transition in this space. The pace of updates from bot management vendors has accelerated, and the old methods of maximizing data success are becoming less effective.

This is a good thing.

The complexity of modern anti-bot measures has gone past the point where human developers can, or should, manage it manually.

...

The friction of the past 15 years is giving way to a future where data extraction is seamless, compliant, and powered by intelligent agents. (5-6) ⁵

For this, and undoubtedly many, companies, the increasing automation is an opportunity to ramp up production. The confident, non-nuanced tone of this report, with turns of phrase like "This is a good thing," and "can, or should", is reminiscent of text generated by an LLM. And why shouldn't it be?

Indeed, AI and data appear to feed one another. In this case, the company focused in gathering data is pushing the AI tools as an efficiency-enhancing component. In turn, the data which is scraped is used to train more models. The data that is scraped by AI will be fed to AI—a kind of digital ouroboros.

Unlike the traditional ouroboros, which is a symbol of eternal and cyclical renewal, this is a cycle whose process of consumption depletes itself.

First, and most obviously, the cycle depletes natural and labor resources. Processing data at this scale requires large scale data processing infrastructures, like data centers. Not only do these centers

Study on Re-Use of the Open Web for AI Training. Zenodo, 2025, p. 75. HAL Archives Ouvertes, <https://doi.org/10.5281/zenodo.15097649>.

⁵Zyte. 2026 Web Scraping Industry Report - PDF. <https://www.zyte.com/whitepaper-ebook/2026-web-scraping-industry-report/>. Accessed 8 May 2026.

exhaust precious water in order to cool them, their disproportionate use of energy most visibly results in rising prices for electricity across the country, and perhaps less visibly (though more devastating) in its effects on the environment (CITATION). And this is just at the higher levels of the processing stack. Acquiring those amounts of hardware and data in the first place requires invasive and exploitative mineral mining practices and data theft at a scale that has never been seen before (CITATION). Indeed, this ouroboros is a cycle that cannot sustain itself.

Second, it also depletes creative resources. For, there is a limit to the content that human beings can create. As more AI-generated content proliferates on the internet, we approach what Matt Kirschenbaum calls the "textpocalypse," where AI-generated text dominates the content on the internet.⁶ Interestingly, researchers have found that this kind of text poisons model training, creating what Shumailov et al. call "model collapse," in which "a degenerative process... in which the data they [the models] generate end up polluting the training set of the next generation" (Shumailov et al 755).⁷

Significantly, as some researchers in the AI Ethics field point out, the desire for size is counterproductive. Written by a group of major figures in the field, including Meredith Whittaker, who organized the Google employee walkouts back in 2019 (CITATION), and has since been a major advocate for accountability by big tech corporations, the paper, "Hype, Sustainability, and the Price of the Bigger-is-Better Paradigm in AI", dubunks what the authors call the "bigger-is-better paradigm" that drives ML development. The chase after ever increasing model sizes, the authors argue, is outpacing hardware development, data curation practices and, most importantly for their argument, performance. Their experiments show that models, after a certain size, show limited gains in model performance. This performance plateaus or even decreases across various types of models, including text-, audio-, and vision-based models. This research builds on the touchstone article, "The Dangers of Stochastic Parrots: Can Language Models Be Too Big?," which was presented at the same conference in 2021, the year be-

⁶Kirschenbaum, Matthew. "Prepare for the Textpocalypse." The Atlantic, 8 Mar. 2023, <https://www.theatlantic.com/technology/archive/2023/03/ai-chatgpt-writing-language-models/673318/>. Technology.

⁷Shumailov, Ilia, et al. "AI Models Collapse When Trained on Recursively Generated Data." *Nature*, vol. 631, no. 8022, July 2024, pp. 755–59. www.nature.com, <https://doi.org/10.1038/s41586-024-07566-y>.

fore ChatGPT was first released. While that article posed the problem of model size as a question—"can language models be too big?"—this one, written in an economy saturated with over-investment, gives an answer: yes, language models can be too big, too big for us to manage. What the authors call "documentation debt," a reference to Bender et al.'s paper, has ballooned to unsustainable levels: as datasets increase in size, it becomes impossible to document them. Additionally, models created from those datasets inadvertently are absorbed into other models trained on top of them, perpetuating the problem to a point where the debt will be truly unpayable. In addition to documentation debt, large models also centralize power, because only the richest tech companies can afford to buy the infrastructure needed to train the models. As a result, the interests of those companies will drive further research in development, which will inevitably be focused on size.

Like with the Zyte report, there is a cycle here in which increasing data and AI technology feed each other. This cycle manifests in real circular relationships between companies, where AI companies who have the technology must work with cloud companies who have the necessary servers and GPUs to handle higher compute. The vast majority of the capital raised by these AI companies are paid to the cloud providers, who in turn are responsible for majority of the investment into the AI companies. As the authors point out, "These circular investment deals can blur the boundary between 'investment' and 'acquisition'" (Varoquaux et al 8).

It is a cycle, as preciently pointed out by Shoshana Zuboff in 2015, that thrives on data's status as what she calls an "electronic text." Zuboff, whose 2015 essay "'Big Other: Surveillance Capitalism and the Prospects of an Information Civilization'" (and followup 2018 book, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*) helped to formalize the notion of capture by captial as the "logic of accumulation" (78).⁸ Her essay analyzes papers written by Google's Chief Economist Hal Varian, where he discusses the potential of "computer-mediated economic transactions" (Rpt. Zuboff 78). Varian identifies four new kinds of business activities made possible by computation, which are "data extraction and

⁸Zuboff's ideas have been percolating for some time. Philip Agre's 1994 paper, "Surveillance And Capture: Two Models Of Privacy," details similar views on computation and surveillance, as well as concepts, such as Agre's "grammar of action," by which a human activity is made computable.

analysis," "new contractual forms," "personalization and customization," and "continuous experiments" (Zuboff 78). Most of these rely on the nearly infinite streams of data that platforms can capture from users, and on the fact that these streams are mostly hidden from view, thus creating asymmetrical relationships between the consumer and the business, which obviates the role of trust. As Zuboff explains, "New possibilities of subjugation are produced as this innovative institutional logic thrives on unexpected and illegible mechanisms of extraction and control that exile persons from their own behavior" (85).

"Surveillance" capitalism is not as new as Zuboff would have it. She describes it alternately as "A new economic order," "A rogue mutation of capitalism," "a new instrumentarian power" (from book, *The Age of Surveillance Capitalism*). As critics like Simone Brown and Ruha Benjamin point out, however, the surveillance logic has existed for centuries. Indeed, slavery, as an economic order, could be described in the same terms that Zuboff uses to describe surveillance capitalism: "A parasitic economic logic in which the production of goods and services is subordinated to a new global architecture of behavioral modification" (from book, *The Age of Surveillance Capitalism*). Brown's work surfaces the quantifying and policing practices that Black people have endured since slave ships. More recently, Benjamin shows how new technologies perpetuate older patterns under the guise of technical values like "neutrality." While "tech advances are sold as morally superior," as Benjamin points out, "they could not exist without data produced through histories of exclusion and discrimination" (5). Values such as "objectivity, efficiency, profitability, and progress" work to effectively siphon the creator from these legacies perpetuated in the product (Benjamin 7). She explains that "tech designers encode judgments into technical systems but claim that the racist results of their designs are entirely exterior to the encoding process" (Benjamin 11-12).

While surveillance capitalism is certainly not new, Zuboff does point out something else, an emergent effect of surveillance.

when it comes to information technology, automation simultaneously generates information that provides a deeper level of transparency to activities that had been either partially or completely opaque. It not only imposes information (in the form of programmed instructions), but it also

produces information... This produces action linked to a reflexive voice, as computer mediation symbolically renders events, objects, and processes that become visible, knowable, and shareable in a new way. (76)

Zuboff uses the term "informate" to describe this tendency of computer mediated action to create byproducts of information that were previously inaccessible, even unimaginable. According to Zuboff, this tendency turns all digital data "electronic text"—text which is ripe for extraction via reading. When computers automate action, they create new texts, like foam rising from a boiling pot.

2. the legal debate over fair use

1.2.4 co-option of language

1. word meaning being co-opted: "freedom" to extract data On January 23, the White House released an Executive Order meant to clear the obstacles preventing technological "innovation" regarding AI development. The goal of the order is "to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security," the order argues, the United States "must develop AI systems that are free from ideological bias or engineered social agendas" (The White House, "Removing Barriers"). The Order called for an "Artificial Intelligence Action Plan," and the White House Office of Science and Technology Policy (OSTP) solicited recommendations from the public.

On March 13, 2025, OpenAI responded to the proposal to the OSTP, offering a set of "recoommendations" for developing AI policy. All of the points in the recommendations are framed via positioning against a common enemy, the People's Republic of China (PRC). References to "American-led AI" adopts language of American values like "freedom" and "democracy," which are posed against PRC's "autocratic, authoritarian AI." First, the recommendations assert a "freedom of intelligence," defined as "the freedom to access and benefit from AGI," presumably meaning consumer access to commercial products, like ChatGPT (Lehane 2). Then, it claims the "freedom to innovate," whereby American companies would be exempt from complying with "overly burdensome state laws" around AI use, such as those requiring safety and bias checks ("AI Laws and Regulation in the United

States (2026)"). Finally, there is a "freedom to learn," which is, rather slyly, the freedom of machines (rather than human beings) to learn. Here the proposal is referring to the protection of copyrighted material, which hasn't been a practical issue for AI development (companies like OpenAI are already indiscriminately scraping the web to train their models), but is a legal issue currently being arbitrated in courthouses across the US (<https://chatgptiseatingtheworld.com/aicopyrightcasetracker/>). The freedom to learn, as the proposal itself admits, is really the "ability to learn from copyrighted material" without having to pay for the content (Lehane 5). Like the freedom to innovate, this freedom also derives its argumentative momentum from China's threatening shadow.

The PRC is unlikely to respect the IP regimes of any of such nations for the training of its AI systems, but already likely has access to all the same data, putting American AI labs at a comparative disadvantage while gaining little in the way of protections for the original IP creators. (page 4)

This framing is not quite faithful to the realities on the ground: the US has, already, disregarded "IP regimes" to get access to "all the same data." However, the remarkable hypocrisy around data theft comes into the starkest illustration when discussing the "fair use" exemption from copyright law.

The rapid advances seen with the PRC's DeepSeek, among other recent developments, show that America's lead on frontier AI is far from guaranteed. Given concerted state support for critical industries and infrastructure projects, there's little doubt that the PRC's AI developers will enjoy unfettered access to data—including copyrighted data—that will improve their models. If the PRC's developers have unfettered access to data and American companies are left without fair use access, the race for AI is effectively over. America loses, as does the success of democratic AI. (9-10)

While the US's access to the same data is "fair," China's access to the same data is "unfettered"—despite there being no difference in

the datasets (something like the entire web) and the methods of web scraping and web crawling, which are used to obtain them.

"Freedom" is strange in this context — it's a US value, being contrasted to China's "autocratic" values. Yet, freedom is deployed in the service of making the US policy effectively more like that of China's. The autocratic qualities of the PRC, which allows it to develop its AI free of regulation and copyright considerations, is exactly what the US is asking for in this proposal.

1.2.5 scraping congress.gov

Scraping the congress.gov servers by constructing a URL using metadata from congressional records. Manipulating the URL to circumventing the protections on the servers for accessing the records.

- the protections on congress.gov
- how to construct a URL
- the dataset - congressional records on anti-trans legislation
- analysis of the dataset? close reading?